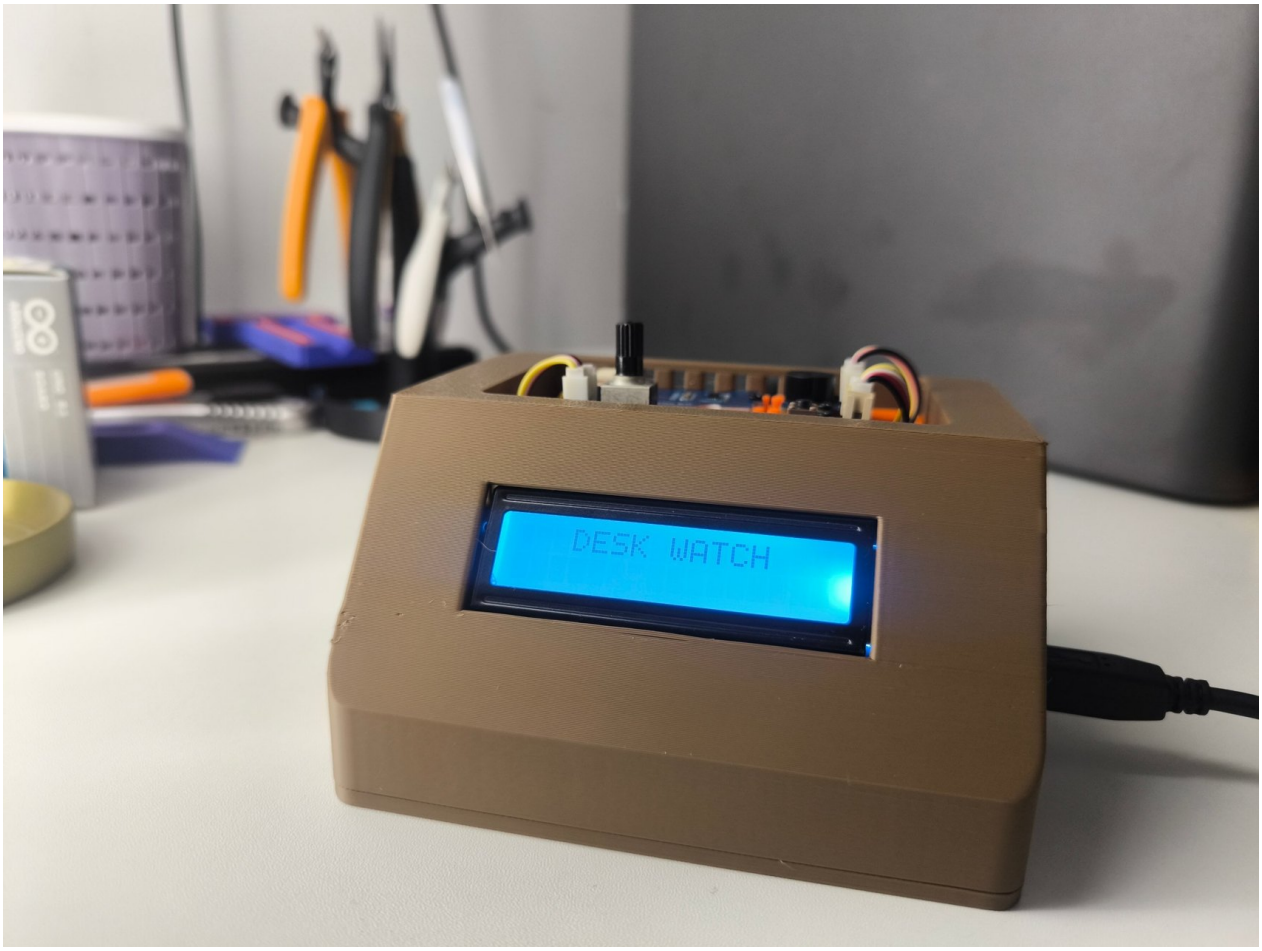
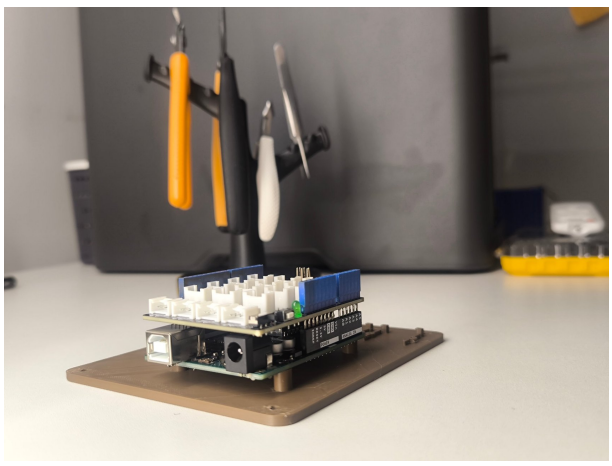
Desk Watch 1

User guide and build reference – MELODY & TEACHER Editions

Arduino UNO + Grove desk console | Pr. BELAMINE MakerLab

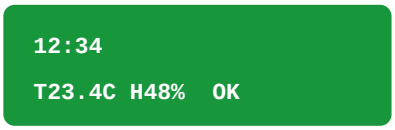






Printed prototype: start-up screen, open top plate with the four Grove modules, and the Arduino UNO with its Base Shield on the base.

Desk Watch 1 is a compact desk console built around an Arduino UNO and a Grove Base Shield. It shows temperature and humidity on a 16x2 colour LCD, warns you when the room goes out of range, and doubles as a timer, stopwatch, hand counter, alarm, dice, chess clock and melody player. Everything is driven by **one button and one dial**, and every module plugs in with a Grove cable: **no soldering**. The **TEACHER Edition** adds an **EXAM** app: an exam countdown and paper collection on one screen.



Everything in range



Temperature too high



Humidity too low

At a glance

- **Climate monitor:** SHT31 sensor, adjustable HIGH / LOW thresholds, colour and buzzer alerts. Each sensor reading is checked (CRC-8); a corrupted reading is ignored.
- **Tools:** Pomodoro timer, stopwatch, hand counter, alarm, dice (D2 to D100), two-player chess clock, Melody player (Happy Birthday, Für Elise, Super Mario Bros, Tetris).
- **Clock and alarm** with the optional RTC module; night mode and automatic sleep.
- **Two editions, two language builds each:** MELODY (10 menu entries) and TEACHER (11 entries, adds EXAM). Each edition ships as an FR (French) and an EN (English) build. The language is fixed by the build and is not a setting; other languages will be created on request in the comments of the model page.
- **Fits an Arduino UNO:** about 74 % of the program memory for MELODY and about 80 % for TEACHER (roughly 6 KB free), 22–23 % of the RAM, with no extra library besides the Grove LCD one.

Status of this release: the four sketches compile for the Arduino UNO. The EXAM app and the latest changes (one language per build, SHT31 CRC check, lighter dice generator) still need a physical device check. Enclosure files are listed on the model page.

1. What you need

Part	Notes
Arduino UNO R3	Any UNO R3 compatible board
Grove Base Shield for Arduino	Keep the voltage switch on 5 V
Grove LCD RGB Backlight V4.0 (16x2)	I2C. Uses the *Grove - LCD RGB Backlight* library (rgb_lcd.h)
Grove SHT31 temperature and humidity sensor	I2C, address 0x44
Grove Buzzer	Digital port D3
Grove Button	Digital port D4
Grove Rotary Angle Sensor (the dial)	Analog port A1
Grove RTC (DS1307 or DS3231), optional	I2C, address 0x68. Needed for the clock, the alarm and the automatic night mode
Grove cables, USB-B cable, small screws	See the assembly guide for the screw plan

2. Wiring

Plug each Grove cable into the matching port of the Base Shield. The three I2C devices (LCD, SHT31 and the optional RTC) can share any I2C ports: they are wired in parallel.

Module	Base Shield port	Signal
LCD RGB Backlight	I2C	SDA / SCL
SHT31	I2C	SDA / SCL (address 0x44)
RTC (optional)	I2C	SDA / SCL (address 0x68)
Buzzer	D3	Digital output, software square wave
Button	D4	Digital input, HIGH when pressed
Rotary Angle Sensor	A1	Analog input, 0 to 1023

The small L LED on the Arduino board (pin 13) lights up while the Desk Watch is in standby, and while you hold the button long enough to put it to sleep. Its colour depends on the board (often orange). To get a red indicator, plug a red Grove LED on a free port and change PIN_LED at the top of the sketch.

3. Installing the firmware

Step	What to do
1	Install the Arduino IDE (2.x), then in the Library Manager install **Grove - LCD RGB Backlight** (Seeed Studio). No other library is needed: the SHT31 and the RTC are read directly over I2C.
2	Choose your edition and language build: `Desk_Watch_1_MELODY_Edition_EN.ino`, `_MELODY_Edition_FR.ino`, `Desk_Watch_1_TEACHER_Edition_EN.ino` or `_TEACHER_Edition_FR.ino`. Keep each file inside its identically named folder, open it, choose **Arduino UNO** and your port, then upload.
3	At power-up you see DESK WATCH, a short melody, then PR BELAMINE MAKERLAB, the edition loading bar and a module check (RTC and sensor).
4	If the dial turns the wrong way for your wiring, invert it in `potRead()`; if your dial is on another port, change `PIN_POT`.

4. Controls

The Desk Watch has one button and one dial. The same gestures work on every page.

Gesture	What it does
Short press	Open the menu from the home screen; select the entry in the menu; confirm a setting. A short press is confirmed 0.3 s after release (the double-click delay).
Double click	Page action: reset the timer or stopwatch, set the thresholds on Temp & Hum, arm or disarm the alarm, pause the chess clock, back to the menu from several pages.
Long press (0.8 to 2.5 s)	Open the menu from any page; inside the menu, go back to the previous page. Timers and the stopwatch keep running in the background. During an EXAM session it opens the exam controls.
Very long press (over 2.5 s)	Standby: the screen switches off and the L LED lights up. Any button press wakes it. The alarm stays active.
Dial	In the menu, scrolls the entries. On a page, chooses a value (timer length, dice sides, chess time control, thresholds...). Turning right moves to the next entry or a larger value.

The menu



MELODY, first entry



TEACHER, first entry



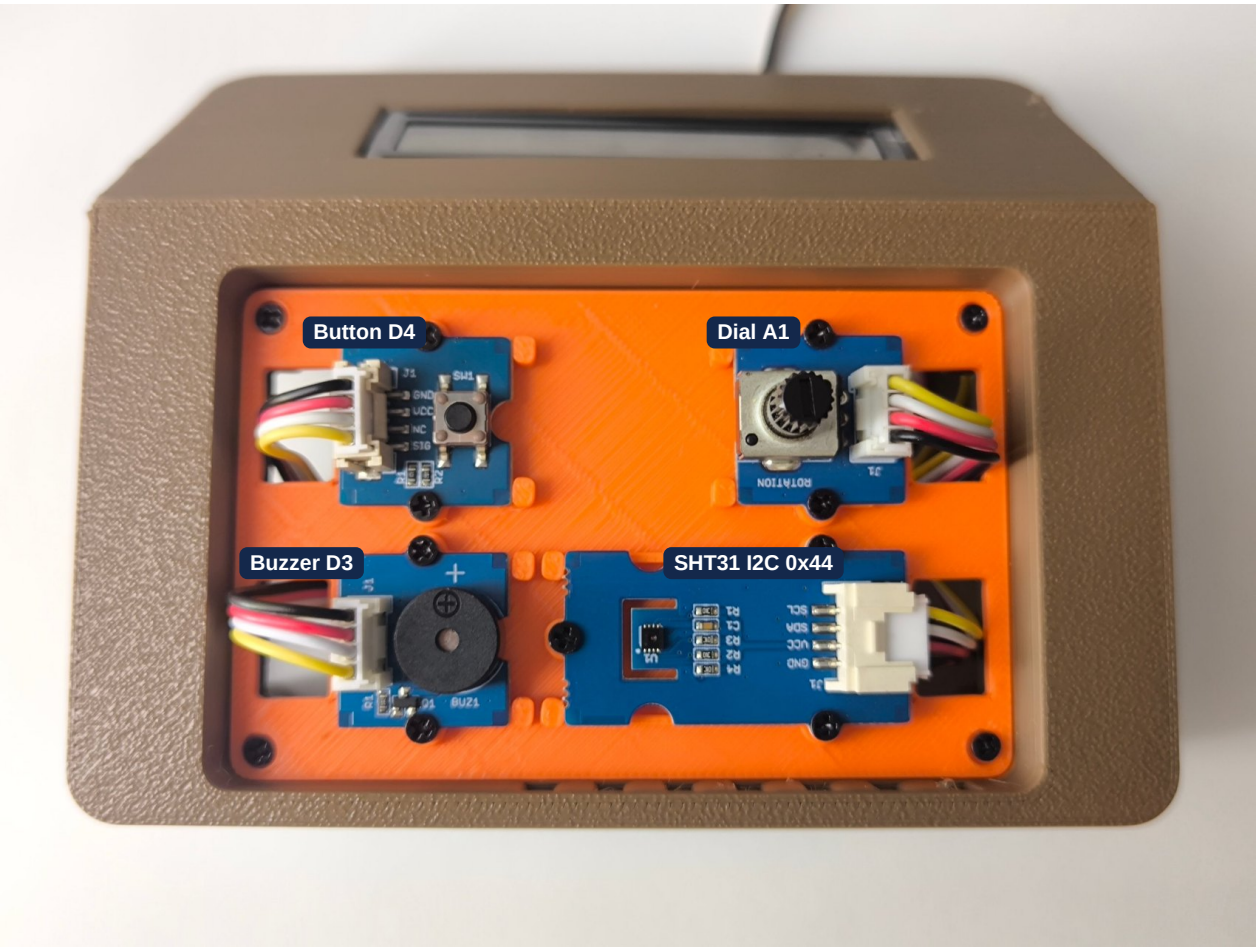
Settings list

Entry	What it opens
Favorite	The page you chose in Settings (Auto by default)
Examen (TEACHER only)	Exam countdown and paper collection
Temp & Hum	Clock, temperature, humidity and alerts
Timer	Pomodoro timer, 1 min to 4 h
Counter	Hand tally counter (shows the running timer or stopwatch)
Stopwatch	Chronograph
Alarm	Set and arm the alarm (needs the RTC)
Dice	Dice from 2 to 100 sides
Chess clock	Two-player chess clock
Melody	Melody player: four tunes
Settings	Brightness, volume, startup sound, auto sleep, night mode, set time, favorite

Menu order: Favorite, [Examen], Temp & Hum, Timer, Counter, Stopwatch, Alarm, Dice, Chess clock, Melody, Settings. Timer is followed immediately by Counter.

Module layout

Top plate seen from above, the screen being at the top of the picture. Each module is held by two screws and plugs into the Base Shield with a short Grove cable that runs down through the side slots.



Top-right badges

When a function runs in the background, a letter appears in the top-right corner of the screen: **!** a temperature or humidity alert, **A** alarm armed, **C** stopwatch running, **M** timer running (the M blinks off for 1 s every 10 s).

5. The pages

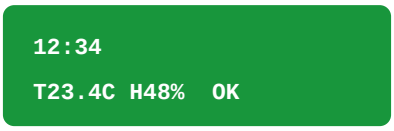
Temp & Hum

The home page of the station. Row 1 shows the RTC time and the background badges; row 2 shows temperature and humidity. Colours tell you the state at a glance: **green** all in range, **red** too high, **blue** too low. When a threshold is crossed the alert alternates with the measurements every 2 s while the clock stays visible. The buzzer gives two short beeps for HIGH and one long beep for LOW (silent in night mode and in standby). A small hysteresis (0.3 C and 2 %) stops the alert from flickering around a threshold.

- **Double click** to set the four thresholds with the dial: temperature low and high (10 to 40 C, steps of 0.5 C), humidity low and high (10 to 90 %). A short press moves to the next one, the last one saves.
- Defaults: 20.0 to 28.0 C and 40 to 60 % humidity. Thresholds are stored and reused at start-up.
- If the sensor is unplugged or returns corrupted data three times in a row, the screen reports the sensor as missing.



Threshold edit



In range



Alert

Timer (Pomodoro)

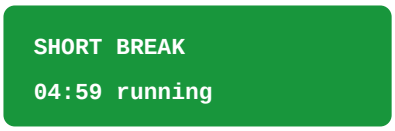
Turn the dial to choose the work time: 1 to 60 minutes in 1-minute steps, then 65 minutes to 4 hours in 5-minute steps. Short press starts or pauses; double click resets. After each work period comes a 5-minute break (a 15-minute break every 4th time). Turn the dial or press long to open the menu: the timer keeps running in the background.



Choosing the length



Running



Break

Stopwatch and Counter

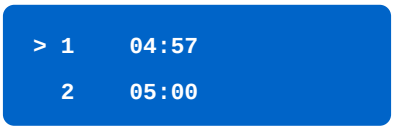
- **Stopwatch:** short press starts or stops, double click resets. Already at zero, a double click or a turn of the dial goes back to the menu.
- **Counter:** every short press adds 1 immediately (up to 9999), even when you click fast. The lower left shows the active timer or stopwatch (they alternate every 2 s if both run). To reset, turn the dial: the screen asks to confirm, a short press means yes, a double click or 5 s without action means no.

Dice, Chess clock and Melody

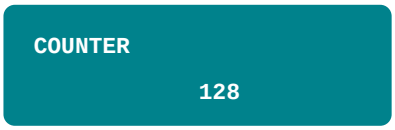
- **Dice:** the dial chooses the number of sides (2, 4, 6, 8, 10, 12, 20, 30 or 100). A short press rolls: the digits scroll with a small tick for about a second, then the result stays with a beep.
- **Chess clock:** the dial chooses the time control (1+0, 3+0, 3+2, 5+0, 5+3, 10+0, 10+5, 15+10, 30+0, 60+0: minutes + seconds added per move). A short press starts with player 1; each press then hands over to the other player. Double click pauses, resets or goes back. Automatic sleep never triggers during a game.
- **Melody:** the dial chooses a tune (Happy Birthday, Für Elise, Super Mario Bros, Tetris), a short press plays, pauses or resumes, a double click goes back to the menu.



Dice



Chess, player 1



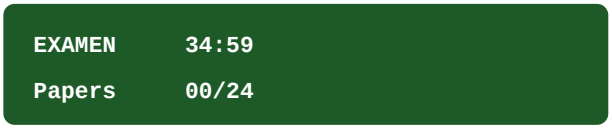
Counter

Alarm

Open **Alarm** in the menu (needs the RTC). A short press steps through hours, minutes, armed, melody and alarm volume; the dial chooses each value and the last short press saves. Melody and volume play a live preview (the melody at minimum volume, the volume at the chosen level). Eight tunes are available: the four Melody tunes plus Aurora, Cosmic Pulse, Arcade Hero and Epic Rise. A double click on the alarm page arms or disarms it. While it rings, a short press or double click snoozes 5 minutes and a long press stops it. The alarm rings even in night mode and in standby.

EXAM (TEACHER Edition)

- Open **Examen** (second menu entry), choose **1–240 minutes** and **1–99 expected papers**; the suggestions are 35 min and 24 papers. A short press then starts the session.
- Every short press counts a paper immediately, even after time is up. A long press (0.8 to 2.5 s) opens the controls: back, subtract one paper, pause or resume, +5 min, end the exam (with confirmation), main menu. The countdown keeps running in the background and automatic sleep is suspended.
- At zero, two short beeps sound once and collection remains available. At the expected total the screen turns green; the final message alternates with the actual elapsed working time, excluding pauses.



Collection



Final summary

6. Settings

Open Settings in the menu. Turn the dial to choose a line, short press to open it, turn the dial to choose the value, short press to confirm. A long press cancels the change and goes back.

#	Setting	Choices	Notes
1	Brightness	8 levels, 5 to 100 %	Applies to the backlight colour of every page; long press cancels
2	Volume	Low / Medium / High	Changes the buzzer duty cycle; a test beep plays while you choose
3	Startup sound	On / Off	Plays a three-note melody at start-up and wake-up
4	Auto sleep	Never, 1, 2, 5, 10, 15, 30 min, 1 h	Screen off after no press and no dial movement. Never by default
5	Night mode	No / Auto 22-6h / Yes	Screen dimmed to 8 %, no alert beeps, no start-up melody. The alarm still rings at full brightness. Auto needs the RTC
6	Set time	Hours, then minutes	Writes the RTC; long press cancels
7	Favorite	Auto, Temp & Hum, Clock, Timer, Stopwatch, Counter	Page opened by Favorite in the menu. Auto alternates the clock and Temp & Hum when an RTC is present; otherwise Temp & Hum only

There is no language setting: each build (FR or EN) uses a single language.

Without the RTC

The station, timers, stopwatch, counter, dice, chess clock and EXAM all work without the RTC. The clock, the alarm and the automatic night mode need it: the screen then asks you to install the RTC module.

7. Troubleshooting

Symptom	Check
Backlight on but no text at start-up	Re-install the Grove LCD library and re-upload; check the LCD cable is on an I2C port.
Sensor missing	SHT31 cable on an I2C port; the sensor must answer at address 0x44. The message also appears after three corrupted readings in a row.
No RTC clock	The RTC module is optional; plug it into an I2C port and restart.
The dial goes the wrong way or jumps	Confirm the dial is on A1 and the shield switch is on 5 V. Invert it in potRead() if needed.
The buzzer is too quiet or too loud	Settings, Volume. Piezo buzzers differ; an active buzzer module ignores the duty cycle.
The screen is almost black at night	Night mode is on (Settings 5). Set it to No, or turn the dial / press the button for the menu.
Compile error: not enough memory	MELODY uses about 74 % and TEACHER about 80 % of the UNO flash. Do not add large features without removing something first.

Desk Watch 1 is part of the Pr. BELAMINE MakerLab range, made by a teacher as a base for classroom electronics projects. Feedback and ideas are welcome in the comments of the model page, including requests for other languages and other editions.